

Derivation of Dual-Rotor Dynamics in Double Spacetime Theory: A Harmonic Oscillator Model from Torsional Mismatch Potential

<https://github.com/hypernumbernet>

Abstract

In the double spacetime theory, every massive particle carries an intrinsic pair of usual and dual spacetimes encoded in the 16-dimensional biquaternion algebra isomorphic to $\text{Cl}(3, 1)$. Gravity arises from the torsional mismatch between the usual rotor R_{usual} and the dual rotor R_{dual} . Here we derive a rigorous dynamical equation for the torsional mismatch angle by promoting the Killing-form invariant J to a particle-intrinsic potential. Variation of the resulting action yields a simple harmonic oscillator equation $\delta\ddot{\theta}_a + m\delta\theta_a = 0$, where m is the rest mass. In the low-energy regime, this naturally reproduces the de Broglie phase, providing a geometric origin for quantum matter waves while preserving classical equivalence to teleparallel gravity.

1 Introduction

The double spacetime theory posits that spacetime is not a shared continuum but a particle-local pair of compactified Minkowski spacetimes described by biquaternions $\mathbb{H} \otimes \mathbb{H} \cong \text{Cl}(3, 1)$. The complete rotor is

$$R_{\text{total}} = R_{\text{usual}} R_{\text{dual}} = \exp \left[\sum_{a=1}^3 \left(\frac{\theta_a}{2} i\Gamma_a + \frac{\phi_a}{2} \Gamma_a \right) \right],$$

where $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$. The classical gravitational action is proportional to the Killing-form scalar J constructed from the bivector $\Omega_{\text{biv}} = \log(R_{\text{usual}}^\dagger R_{\text{dual}})$.

In this paper, we treat J as an intrinsic potential for individual particles and derive the exact dynamics of the torsional mismatch angle $\delta\theta_a = \theta_a - \phi_a$. The resulting equation is a mass-proportional harmonic oscillator, yielding the de Broglie phase in the linear regime.

2 Torsional Mismatch Potential in the Small-Angle Approximation

Define the relative rotor $\Omega = R_{\text{usual}}^\dagger R_{\text{dual}}$. For small mismatch angles,

$$\Omega \approx 1 + \frac{1}{2} \sum_{a=1}^3 \delta\theta_a (i\Gamma_a + \Gamma_a).$$

The bivector $\Omega_{\text{biv}} = \log \Omega$ yields, after cancellation of cross terms and proper normalization,

$$J \approx \frac{1}{2} \sum_{a=1}^3 \delta\theta_a^2.$$

Incorporating the rest mass m as the stiffness of the dual sector,

$$J = \frac{1}{2} m \sum_{a=1}^3 \delta\theta_a^2.$$

3 Particle-Intrinsic Action Principle

We propose an effective action for the intrinsic degrees of freedom of a single particle:

$$S_{\text{particle}} = \int \left[\frac{1}{2} \sum_{a=1}^3 \dot{\theta}_a^2 - \frac{1}{2} \sum_{a=1}^3 \dot{\phi}_a^2 - \frac{m}{2} \sum_{a=1}^3 (\theta_a - \phi_a)^2 \right] dt.$$

The kinetic terms reflect boost freedom in the usual sector and the time-reversed character of the dual sector; the potential is proportional to the torsional mismatch invariant J . Collective averaging over many particles recovers the classical teleparallel action $\int J d^4x$.

4 Derivation of the Equations of Motion

The Lagrangian is

$$L = \frac{1}{2} \sum_{a=1}^3 \dot{\theta}_a^2 - \frac{1}{2} \sum_{a=1}^3 \dot{\phi}_a^2 - \frac{m}{2} \sum_{a=1}^3 (\theta_a - \phi_a)^2.$$

The Euler–Lagrange equations read

$$\ddot{\theta}_a - m(\theta_a - \phi_a) = 0, \tag{1}$$

$$-\ddot{\phi}_a - m(\theta_a - \phi_a) = 0. \tag{2}$$

Introducing the torsional mismatch angle $\delta\theta_a = \theta_a - \phi_a$, we obtain

$$\delta\ddot{\theta}_a + m\delta\theta_a = 0. \tag{3}$$

This is a harmonic oscillator with angular frequency $\sqrt{m}$.

5 Solution for Free Particles and the de Broglie Phase

For a free particle, $\dot{\theta}_a = \gamma v_a/c$ (constant). The general solution of Eq. (3) is

$$\delta\theta_a(t) = A \sin(\sqrt{m}t) + B \cos(\sqrt{m}t).$$

In the low-energy (Compton-scale) regime, rapid oscillations average out, leaving linear accumulation

$$\delta\theta_a(t) \approx \dot{\delta\theta}_a(0) t,$$

with initial rate $\dot{\delta\theta}_a(0) \propto \gamma v_a$. Integrating gives the accumulated phase

$$\int \delta\theta_a dt \propto (\gamma mc^2 t - \gamma m v_a x)/\hbar,$$

which, after transformation to laboratory coordinates, is precisely the de Broglie phase $(Et - \mathbf{p} \cdot \mathbf{x})/\hbar$.

6 Connection to Quantum Effective Theory

Averaging over Compton-frequency oscillations retains only the slowly varying component of $\delta\theta_a$, yielding $\phi_a \approx -\theta_a +$ (slow lag). This linear phase evolution corresponds to the phase $S(x, t)/\hbar$ of the quantum wave function, naturally emerging as the low-energy effective theory (cf. the linearization in Appendix B of the companion paper).

7 Conclusion

By elevating the classical Killing-form invariant J to a particle-intrinsic potential, we have derived the exact dynamical equation $\delta\ddot{\theta}_a + m\delta\theta_a = 0$ from an action principle. This simple harmonic oscillator governs dual-rotor lag, reproduces the de Broglie phase geometrically, and provides the most consistent, background-independent mechanism for the dual-rotor dynamics in double spacetime theory.